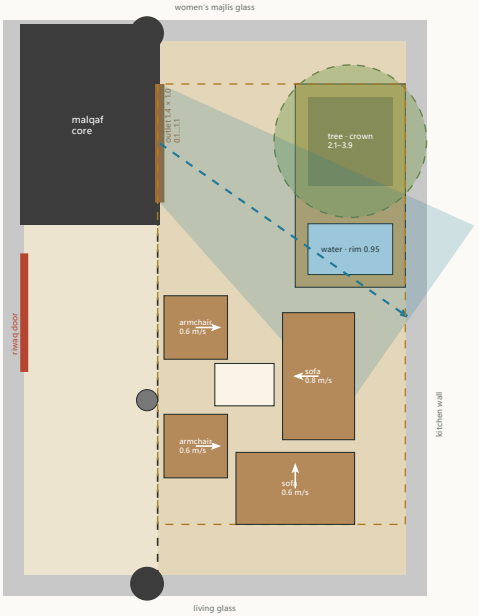
The malqaf outlet turned toward the water · option B against the layout of 10 Nov 2026 (A) · ADOPTED (H2, 10 Nov 2026)

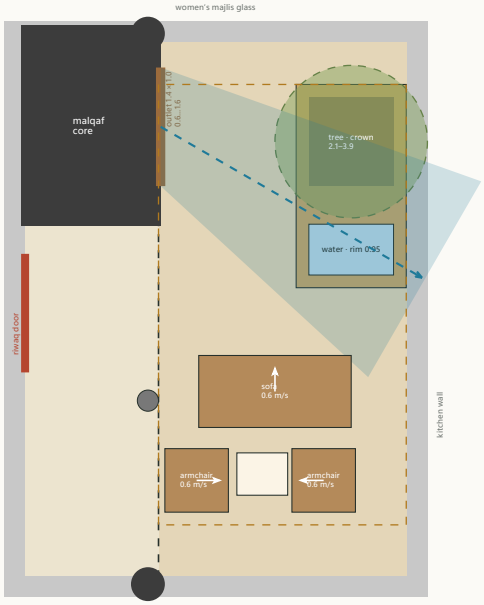
ADOPTED (owner, 10 Nov 2026, with the courtyard layout H2 of studies/courtyard_haven.typ): the outlet 1.4 × 1.0 at y 6.9–8.3, +0.60…+1.60, the louvres fixed 30° south-east; the final seating (a 3-seat sofa facing the water, two armchairs, the olive in its fixed pot) and the water trough are in `studies/courtyard_layout.typ`; the tree and the enclosure drawn here are the earlier ones.

The owner's question (10 Nov 2026): re-orient the outlet toward the tree and water enclosure so the air passes over the water and under the tree before the seating; the seats downstream of the water, every seat with a back, the same clearances; the old outlet face a plain rendered wall flush with the columns. **Where the outlet can go:** the core's north face is the women's majlis wall (nothing can open there); its west face is the riwaq; its south face the covered strip and the door: the courtyard outlet can only be on the **east face**, in the first arcade bay (y 6.65–8.52 between the column faces), as a **turned** outlet: the opening kept at 1.4 × 1.0, **raised** to +0.6…+1.6 so the jet clears the trough's rim (+0.95) and the louvres **fixed** 30° to the south-east toward the trough, the reveals splayed the same way. Velocities at the louvres 2.1 m/s at the summer-evening mean wind (`studies/air_system.typ`; 1 m/s in a gentle wind); the figures at the seats are a free-jet estimate (±30 %), the MEP engineer confirms.

A · 10 Nov 2026: the outlet low (+0.10…+1.10), aimed south-east; 6 seats



B · the outlet raised (+0.60…+1.60) and turned 30° at the trough; 5 seats downstream



Air at the seats (m/s at the summer-evening mean wind; the jet's axis dashed, its spread shaded)

A · seat	from the outlet	≈ m/s
sofa A · 2 seats · faces W: the outlet arch	3.3 m	0.8
sofa B · 2 seats · faces N: the tree and the water	4.4 m	0.6
armchair 1 · faces E	2.2 m	0.6
armchair 2 · faces E	3.6 m	0.6

B · seat	from the outlet	≈ m/s
sofa · 3 seats · faces N: the water 0.8 m ahead, the tree, the outlet beyond	3.4 m	0.6
armchair 1 · faces E, across the table	4.2 m	0.6
armchair 2 · faces W	4.6 m	0.6

Clearances (both keep them)

Path	A	B
colonnade walkway: nothing west of the soffit edge (x 5.93)	0.08 east ✓	0.08 east ✓
at the living glass ≥ 0.60	0.6	0.75
at the women's majlis glass ≥ 0.50	3	3.7
at the kitchen wall (cleaning) ≥ 0.60	0.6	0.6
seats · every one with a back	6 (2 + 2 + 1 + 1)	5 (3 + 1 + 1)

The view from each sofa

- **A · sofa A** (faces west): the outlet arch, the core's rendered face and, above, the blind arch and the malqaf head against the sky; the tree at its right shoulder. **A · sofa B** (faces north): the tree and the water straight ahead, the women's majlis glass beyond, the outlet arch at its left. **The armchairs** (face east): the enclosure and the kitchen wall.
- **B · the sofa** (faces north, 0.8 m from the trough): the water and the lip straight ahead, the tree's crown over it, the outlet arch at the left; nobody looks at the core's face. **The armchairs** face each other across the table, the water in the corner of the eye; the second sofa of A does not fit downstream without looking at the first one's back, so B seats **five**.

The air path in B (the section on page 2)

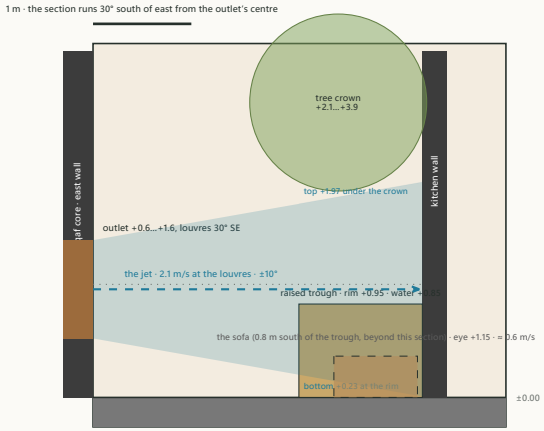
- From the raised outlet (+0.6…+1.6) the jet runs 30° south of east: at the trough's near rim (2.1 m) it spans +0.23…+1.97: **59 % of its depth passes over the water** (rim +0.95, water +0.85), the rest meets the trough's side; its top stays **under the tree's crown** (bottom +2.1) and through the crown's south-west edge; it reaches the kitchen wall beside the trough, spreads along the wall and settles on the sofa 0.8 m south of the water: ≈ 0.6 m/s there, a little cooler and moister for the 1 m of water (≈ −0.5 K, assumed) and with the tree's scent.
- In A the jet leaves low (+0.10…+1.10) and runs south-east over the floor straight to sofa A and the armchairs (≈ 0.8 m/s at sofa A); it misses the water and the tree.

What B costs · verdict for the owner

- **The outlet:** raised 0.5 m and centred in the bay (the air study, the 3D, the DXF, the facade of the core: the arch reads better with the opening centred); the dust trap and the louvres' reach unchanged. **The old low opening** closed: the face a plain rendered wall flush with the columns.
- **Seats:** 5 instead of 6, every one with a back, all downstream of the water; a sixth seat only as a stool or a second sofa behind the first.
- **Air:** A gives the stronger breeze at the seats (direct); B gives a gentler, cooler, scented breeze over the water, the family facing the water and the tree. Both keep the clearances.
- **Recommendation:** if the owner values the water and the tree in front of the seats more than the sixth seat and the stronger direct jet, take B; otherwise keep A. Either way the outlet should be centred in the bay.

Option B study checks passed (3)

Option B · the air path from the outlet over the water and under the tree to the sofa · section along the jet’s axis



What the section shows

- The raised outlet puts the jet’s **lower edge** at +0.23 and its **upper edge** at +1.97 where it meets the trough’s near rim (2.1 m from the louvres): the water (+0.85) lies under the jet’s upper 59 %; the lower part strikes the trough’s side and curls over it. The crown’s bottom (+2.1) stays above the jet: **the air goes under the tree and over the water, then the kitchen wall turns it down and south** onto the sofa 0.8 m beyond.
- The louvres fixed at 30° (the owner can still close them with the switch; the damper D2 behind them is unchanged); the splayed reveals keep the opening’s 1.4 m width square to the arch and give the jet its direction without a visible duct.
- **If A is kept:** only the raising / centring is worth taking (the opening centred in the bay, +0.60…+1.60 or +0.10…+1.10 as the owner prefers: the lower outlet gives the stronger breeze at ankle-to-knee height, the higher one at the body).

If B is adopted (the next release)

`studies/air_system.typ` (`air-outlet` → (5.8, 6.9, 8.3, 0.6, 1.6), the louvres fixed 30° SE, the splayed reveals), `studies/courtyard_layout.typ` (the furniture B, 5 seats), `design_gf.typ` and the presentation’s sheet 02 (the outlet line, the seating), `3d/params.yaml` (the outlet window), the 3D and the DXFs, `PROJECT_V10.md`; the MEP engineer confirms the jet. Nothing of this is applied now.

الخيار B (دراسة، غير معتمد): لا يمكن فتح مخرج الملقف في وجه النواة الشمالي (جدار مجلس النساء) ولا الغربي (الرواق) ولا الجنوبي (الممر والباب)؛ المخرج يبقى في الوجه الشرقي في أول قوس، لكنه **يُرفع** إلى +0.6…+1.6 ويُوجَّه بشيش ثابت وعضادات مائلة 30° نحو حوض الماء، فيمر الهواء فوق الماء (نحو 59 % من عمق التيار فوق حافة الحوض) وتحت تاج الشجرة ثم يرتد عن جدار المطبخ إلى الجلسة. **الجلسة** تصبح خلف الماء: كنبة بثلاثة مقاعد تواجه الماء والشجرة وكرسيان متقابلان حول الطاولة: خمسة مقاعد كلها بظهر (السادس لا يتسع إلا خلف الكنبة)؛ المسافات الخالية نفسها. الفتحة القديمة المنخفضة تُغلق ويصبح الوجه لياسة ملساء على خط الأعمدة. **المقارنة:** A نسيم أقوى ومباشر وستة مقاعد؛ B نسيم ألطف وأبرد قليلا فوق الماء والعائلة تواجه الماء والشجرة بخمسة مقاعد. لا يُطبّق شيء قبل موافقة المالك.